

This is a classic bond pricing problem. When a bond's coupon rate is exactly equal to its required rate of return (or market interest rate), it will always sell at its **par value** (face value).

Since a standard face value for corporate bonds is typically **\$1,000** (unless stated otherwise), the value of this bond will be exactly **\$1,000**.

Here is the deep-dive explanation, the formula, and the step-by-step mathematical proof of why this happens.

1. Understanding the Concept

To value a bond, you need to calculate the **Present Value (PV)** of all the cash flows it will generate in the future. A standard coupon bond provides two types of cash flows:

1. **Annual Coupon Payments:** A stream of fixed annuity payments paid every year until maturity.
2. **Maturity Value (Par Value):** A one-time lump-sum payment returned to the investor at the end of the 10 years.

Given Data:

- **Maturity (N or n):** 10 years
 - **Coupon Rate:** 10%
 - **Face Value (M or F V):** \$1,000 (Standard default)
 - **Annual Coupon Payment (INT):** $10\% \times \$1,000 = \100 per year
 - **Required Return (r_d):** 10% (0.10 as a decimal)
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2. The Bond Valuation Formula

The total value of the bond (V_B) is the sum of the present value of the coupons (using the annuity formula) and the present value of the face value (using the lump-sum formula):

$$V_B = INT \times \left[\frac{1 - (1 + r_d)^{-n}}{r_d} \right] + \frac{M}{(1 + r_d)^n}$$

Where:

- V_B = Value of the bond
 - INT = Annual coupon payment amount (\$100)
 - r_d = Market interest rate / required rate of return (10%)
 - n = Number of years to maturity (10)
 - M = Maturity value or Face value (\$1,000)
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3. Step-by-Step Solution

Let's plug the values into the formula to prove the math:

$$V_B = 100 \times \left[\frac{1 - (1 + 0.10)^{-10}}{0.10} \right] + \frac{1,000}{(1 + 0.10)^{10}}$$

Part A: Present Value of Coupon Payments (Annuity)

1. Calculate the discount factor: $(1.10)^{10} \approx 2.59374$
2. Find the inverse: $(1.10)^{-10} = \frac{1}{2.59374} \approx 0.38554$
3. Compute the numerator: $1 - 0.38554 = 0.61446$
4. Divide by r_d : $\frac{0.61446}{0.10} = 6.1446$ (This is your PVIFA factor)
5. Multiply by the coupon: $\$100 \times 6.1446 = \614.46

Part B: Present Value of the Face Value (Lump Sum)

1. Divide the face value by the discount factor: $\frac{1,000}{(1.10)^{10}} = \frac{1,000}{2.59374} = \385.54

Part C: Combine the Two Values

$$\$V_B = \$614.46 + \$385.54 = \mathbf{\$1,000.00}$$

The Core Takeaway: The Par Bond Rule

This problem illustrates a fundamental rule in fixed-income finance:

- **Par Bond** ($r_d = \text{Coupon Rate}$): When the market interest rate equals the coupon rate, the bond sells at **Par** (Value = Face Value).
- **Discount Bond** ($r_d > \text{Coupon Rate}$): If the market rate rises above 10%, this bond's 10% payout becomes less attractive, and its price will drop **below** \$1,000.

- **Premium Bond ($r_d < \text{Coupon Rate}$):** If the market rate drops below 10%, this bond's 10% payout becomes highly attractive, and its price will rise **above** \$1,000.